

C955

Applied Probability and Statistics

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

Method © 2026 Erik Daniel Lindquist — Course Mastery Coach, armored drill. Items drafted with AI assistance under his direction.

How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A data center in Phoenix logs server response times. The first quartile is 20 ms, the median is 29 ms, and the third quartile is 40 ms. Using the $1.5 \times \text{IQR}$ rule, which of these response times is an outlier?

- A. 72 ms
- B. 45 ms
- C. 12 ms
- D. 65 ms

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

A data set has $Q1 = 20$ and a lower outlier fence of 5. What is the IQR?

- A. 20
- B. 10
- C. 15
- D. 22.5

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A help desk that has used the same ticketing system since 2022 receives a bell-shaped number of tickets each month, with a mean of 400 and a standard deviation of 30. About what percent of months have between 340 and 460 tickets?

- A. 47.5%
- B. 99.7%
- C. 95%
- D. 68%

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

In a bell-shaped distribution, about 68% of values fall between 85 and 115. What are the mean and standard deviation?

- A.** Mean 100, SD 30
- B.** Mean 100, SD 15
- C.** Mean 115, SD 15
- D.** Mean 100, SD 7.5

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

Using 36 months of data from four stores, an analyst fits predicted sales (\$ thousands) = 12 + 3.5 × ad spending (\$ thousands), with $r^2 = 0.64$. The marketing team is based in Austin. What does the slope mean?

- A.** Each \$3,500 in ads adds \$1,000 in sales.
- B.** With zero ad spending, predicted sales are \$3,500.
- C.** Each \$1,000 in ads causes exactly \$3,500 more in sales.
- D.** Each additional \$1,000 in ads is associated with about \$3,500 more in predicted sales.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

An IT director finds $r = 0.70$ between the number of chat messages a project team sends and its project success score. The analysis covered 48 teams over two fiscal years. She proposes requiring every team to send more messages. What is the best assessment?

- A.** Any r below 1.0 means there is no relationship.
- B.** Mandate it; $r = 0.70$ proves messaging drives success.
- C.** The link may reflect something else, such as team engagement; test the policy before mandating it.
- D.** 70% of project success is due to messaging.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

An order system needs both a web server and a database server to be up. The servers are independent, sit in separate racks, and each cost \$8,000. Each is available 99% of the time. What is the probability the order system is up?

- A. 0.9999
- B. 1.98
- C. 0.99
- D. 0.9801

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

A company has two independent backup systems, and each fails 10% of the time. What is the probability that at least one backup works?

- A. 0.99
- B. 0.81
- C. 0.01
- D. 0.90

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

Of 500 customers, 200 use the company's mobile app, which launched in 2023 and has a 4.6-star rating. Of the app users, 120 renewed their subscription; of the 300 non-users, 150 renewed. Are app use and renewal independent?

- A. Yes, because 120 is less than 150.
- B. Yes, since both groups have renewals.
- C. No, because $P(\text{app}) = 0.40$ is not 0.54.
- D. No; $P(\text{renew} | \text{app}) = 0.60$ differs from $P(\text{renew}) = 0.54$.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A company with four regions of very different sizes wants to survey employees. HR, which licensed a new survey platform last year, randomly selects employees within every region in proportion to its size. Which sampling method is this?

- A.** Stratified sampling
- B.** Systematic sampling
- C.** Simple random sampling
- D.** Cluster sampling

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A 45-person company that switched cloud providers in month 1 had a bill of \$2,400 in month 3 and \$3,300 in month 9. The bill is growing linearly. What is the monthly rate of change?

- A.** About \$366.67 per month
- B.** \$900 per month
- C.** \$150 per month
- D.** About \$0.0067 per month

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12. [TRANSFER — reversed stem / different variable]

A cloud bill is \$2,400 in month 3 and grows by \$150 per month. In what month will it reach \$4,200?

- A.** Month 9
- B.** Month 15
- C.** Month 28
- D.** Month 12

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer A Concept 1

A data center in Phoenix logs server response times. The first quartile is 20 ms, ~~the median is 29 ms~~, and the third quartile is 40 ms. Using the $1.5 \times \text{IQR}$ rule, which of these response times is an outlier?

Target word: which value is an outlier **Decides:** Upper fence = $40 + 1.5 \times 20 = 70$; lower fence = -10 .

A	Correct	Above the upper fence of 70 ms.
B	Adjacent category	Above Q3 is not the same as an outlier.
C	Inverted operation	Low but above the lower fence of -10 .
D	Dropped element	Above $Q3 + \text{IQR} = 60$ but below 70; drops the 1.5.

2. Answer B Concept 1 • transfer

A data set has $Q1 = 20$ and a lower outlier fence of 5. What is the IQR?

Target word: solve for IQR **Decides:** $20 - 1.5 \times \text{IQR} = 5$.

A	Wrong stem number	Q1 itself.
B	Correct	$15 / 1.5 = 10$.
C	Dropped element	Forgets the 1.5 multiplier.
D	Inverted operation	Multiplies 15 by 1.5 instead of dividing.

3. Answer C Concept 2

A help desk ~~that has used the same ticketing system since 2022~~ receives a bell-shaped number of tickets each month, with a mean of 400 and a standard deviation of 30. About what percent of months have between 340 and 460 tickets?

Target word: percent within a range **Decides:** 340 to 460 is ± 2 SD.

A	Dropped element	Only one side of the mean.
B	Adjacent category	That is ± 3 SD, 310 to 490.
C	Correct	± 60 is 2 SDs; about 95%.
D	Confusion set	That is ± 1 SD, 370 to 430.

4. Answer B Concept 2 • transfer

In a bell-shaped distribution, about 68% of values fall between 85 and 115. What are the mean and standard deviation?

Target word: solve for mean and SD **Decides:** 68% is ± 1 SD, so the SD is half the width.

A	Dropped element	Uses the full width without halving.
B	Correct	Midpoint 100; half-width 15 = 1 SD.
C	Wrong stem number	Uses the upper bound as the mean.
D	Confusion set	Treats the interval as ± 2 SD, the 95% rule.

5. Answer D Concept 3

Using 36 months of data from four stores, an analyst fits predicted sales (\$ thousands) = $12 + 3.5 \times \text{ad spending} (\$ \text{ thousands})$, with $r^2 = 0.64$. ~~The marketing team is based in Austin.~~ What does the slope mean?

Target word: slope interpretation **Decides:** $3.5 = \text{change in predicted sales } (\$000) \text{ per } \$1,000 \text{ of ads.}$

A	Inverted operation	Reverses x and y.
B	Confusion set	Mixes up slope and intercept; the intercept is \$12,000.
C	Absolute language	A regression shows association and average prediction, not exact cause.
D	Correct	Slope is the change in y per unit of x, in thousands.

6. Answer C Concept 4

An IT director finds $r = 0.70$ between the number of chat messages a project team sends and its project success score. ~~The analysis covered 48 teams over two fiscal years.~~ She proposes requiring every team to send more messages. What is the best assessment?

Target word: what r supports **Decides:** Correlation, not proof that more messages cause success.

A	Absolute language	0.70 is a fairly strong association.
B	Confusion set	Treats correlation as causation.
C	Correct	A lurking variable could drive both; only an experiment tests cause.
D	Wrong stem number	Misreads r as a causal percent.

7. Answer D Concept 5

An order system needs both a web server and a database server to be up. The servers are independent, ~~sit in separate racks, and each cost \$8,000.~~ Each is available 99% of the time. What is the probability the order system is up?

Target word: P(both up) **Decides:** Both needed: multiply.

A	Confusion set	That is P(at least one up), the backup case.
B	Inverted operation	Adds instead of multiplying; a probability cannot exceed 1.
C	Dropped element	Counts only one server.
D	Correct	$0.99 \times 0.99 = 0.9801$.

8. Answer A Concept 5 • transfer

A company has two independent backup systems, and each fails 10% of the time. What is the probability that at least one backup works?

Target word: P(at least one) **Decides:** $1 - P(\text{both fail}) = 1 - 0.1 \times 0.1$.

A	Correct	$1 - 0.01 = 0.99$.
B	Confusion set	0.9×0.9 is P(both work).
C	Inverted operation	P(both fail), the complement.
D	Dropped element	Counts only one backup.

9. Answer D Concept 6

Of 500 customers, 200 use the company's mobile app, ~~which launched in 2023 and has a 4.6 star rating~~. Of the app users, 120 renewed their subscription; of the 300 non-users, 150 renewed. Are app use and renewal independent?

Target word: independence test **Decides:** Compare $P(\text{renew} \mid \text{app})$ with $P(\text{renew})$.

A	Wrong stem number	Raw counts come from groups of different sizes.
B	Dropped element	Independence needs equal rates, not just renewals in both groups.
C	Adjacent category	Compares two unrelated probabilities.
D	Correct	$120 / 200 = 0.60$; $270 / 500 = 0.54$.

10. Answer A Concept 7

A company with four regions of very different sizes wants to survey employees. HR, ~~which licensed a new survey platform last year~~, randomly selects employees within every region in proportion to its size. Which sampling method is this?

Target word: sampling method **Decides:** A random sample is taken inside every group.

A	Correct	Every region is a stratum sampled at random.
B	Adjacent category	Systematic takes every kth person from a list.
C	Dropped element	Drops the region structure built into the design.
D	Confusion set	Cluster sampling picks whole groups at random and leaves others out.

11. Answer C Concept 8

A ~~45-person~~ company ~~that switched cloud providers in month 1~~ had a bill of \$2,400 in month 3 and \$3,300 in month 9. The bill is growing linearly. What is the monthly rate of change?

Target word: slope **Decides:** $(3,300 - 2,400) / (9 - 3)$.

A	Wrong stem number	$3,300 / 9$ divides one value by its month.
B	Dropped element	Forgets to divide by the 6 months.
C	Correct	$900 / 6 = 150$.
D	Inverted operation	$6 / 900$ flips the ratio.

12. Answer B Concept 8 • transfer

A cloud bill is \$2,400 in month 3 and grows by \$150 per month. In what month will it reach \$4,200?

Target word: solve for x **Decides:** $3 + (4,200 - 2,400) / 150$.

A	Adjacent category	Reuses a month from a different data point.
B	Correct	$1,800 / 150 = 12$ months after month 3.
C	Wrong stem number	$4,200 / 150$ uses the total instead of the increase.
D	Dropped element	Forgets to add the starting month.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	10	
Wrong stem number	A figure copied from the prompt that is not the answer.	6	
Adjacent category	The right idea in the wrong bucket.	5	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	6	
Absolute language	Always, never, guarantee — where judgment applies.	2	
Confusion set	Every option is real; the test is exact role-matching.	7	
Camouflage	Answers a stem fact that was never the question.	0	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	IQR and the 1.5 × IQR outlier rule	$IQR = Q3 - Q1$; values below $Q1 - 1.5 \times IQR$ or above $Q3 + 1.5 \times IQR$ are outliers.	Flagging anything above $Q3$, or leaving out the 1.5.	Quartiles given and a question about unusual values.
2	Empirical rule	In a bell-shaped distribution about 68%, 95% and 99.7% of values fall within 1, 2 and 3 SDs of the mean.	Matching the wrong percent to the number of SDs.	“Bell-shaped” or “normal” plus a range around the mean.
3	Reading a regression slope	The slope is the predicted change in y per one-unit change in x , in the units of y and x ; it describes association, not guaranteed cause.	Mixing up slope and intercept, or reading the slope as a percent.	An equation $\hat{y} = a + bx$ with units attached.
4	Correlation vs. causation	A correlation, however strong, does not show that changing x will change y ; lurking variables and reverse causation are possible.	Acting on r as if it proved cause.	A proposal to mandate or increase x because it correlates with y .
5	Independent events: “and” vs. “at least one”	$P(A \text{ and } B) = P(A) \times P(B)$ for independent events; $P(\text{at least one}) = 1 - P(\text{none})$.	Adding probabilities, or mixing up “both” with “at least one”.	Systems in series (all needed) vs. backups (any one enough).
6	Testing independence	A and B are independent only if $P(A B) = P(A)$.	Comparing unrelated probabilities, or raw counts.	A two-way table and the word “independent”.
7	Sampling designs	Stratified = random sample within every group; cluster = randomly choose whole groups; systematic = every k th item.	Confusing stratified with cluster sampling.	Whether every group is sampled, or only some whole groups.
8	Slope as a rate of change	Slope = change in y / change in x between two points; a linear model grows by the same amount each period.	Forgetting to divide by the change in x , or dividing y by x directly.	Two (time, value) pairs and a question about rate or a future value.

Course scope (public description): Applied Probability and Statistics is designed to help students develop competence in the fundamental concepts of basic statistics including: introductory algebra and graphing; descriptive statistics; regression and correlation; and probability. Statistical data and probability are often used in everyday life, science, business, information technology, and educational settings to make informed decisions about the validity of studies and the effect of data on decisions. This course discusses what constitutes sound research design and how to appropriately model phenomena using statistical data. Additionally, the content covers simple probability calculations, based on events that occur in the business and IT industries. No prerequisites are required for this course.